

Polar Coordinates and Conversions



Converting points

- 1 Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $(4, \frac{\pi}{3})$

b $(2, \frac{3\pi}{4})$

- 2 Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$

b $(0, -5)$

Converting equations

- 3 Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

- 4 Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$

b $y = x$

- 5 Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.